Eulerian hypermap partial duals via crossing-total medial orientations

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Editorial note. Complete original author body, essential flag diagrams and bibliography retained; one selected primary per task. No proof repair, added assumptions or mathematical certification.

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Allowed standard prerequisites (editorial).

- Embedded hypermaps, flag involutions and medial maps. All native flag involutions, edge-local partial duals, state-circle conventions and modified-medial twist data retained with original source diagrams and bibliography.. The selected global orientation-characterization proof and its load-bearing author lemmas remain present in full.

1 Introduction

Partial duality is a local form of geometric duality. Chmutov introduced the notion for ribbon graphs [1], and Chmutov and Vignes-Tourneret subsequently extended it to hypermaps [2]. Structural studies of partial duals of ribbon graphs include the work of Moffatt [17, 18]. For ribbon graphs, Huggett and Moffatt characterized bipartite partial duals of plane graphs by all-crossing orientations of the medial graph [11]. Metsidik and Jin later gave an Eulerian characterization for plane graphs [16]; Deng, Jin, and Metsidik extended both orientation characterizations to arbitrary ribbon graphs [6]. Their work shows that the modified medial graph is indispensable in the non-orientable bipartite case.

Combinatorial models for hypermaps go back to work of Cori, Walsh, Jones, Singerman, and others; background can be found in [3, 20, 13, 14]. The framework encoding maps on surfaces by flags, fixed-point-free involutions, and edge-coloured graphs appears in [15, 19]. Standard background on maps on surfaces,

ribbon graphs, and their ribbon representations is given in [9, 8]. For orientable hypermaps, the permutation model (σ, α) conveniently describes hyperedge partial duality and medial maps; Cori and Hetyei gave a permutation construction of the medial map in that setting [4]. Han and Metsidik established orientation characterizations of Eulerian and bipartite hyperedge partial duals in this model [10, Theorems 5.3 and 6.5]. However, the left–right x/y labels used by the (σ, α)-model and its bipartiteness proof depend on a global orientation of the underlying surface.

Our aim is to formulate both orientation characterizations uniformly for all hypermaps. We therefore use the τ-model, also called the dart or flag model. It records the complete local cell structure and applies equally well on non-orientable surfaces. In particular, we give a direct construction of the medial map from this model and use local orientation systems to record how bands are attached relative to vertex discs.

The two problems have different sensitivities to orientability. The Eulerian characterization is controlled by alternating signs on state circles and requires no global left–right convention. The usual proof of the bipartite characterization does require such a convention. We replace it by local orientation systems on medial edges and their twist data; equivalently, every medial edge whose endpoint local orientations are incompatible is subdivided by an artificial bivalent bar. This is precisely the hypermap analogue of the modified medial map in [6]. Related background on twists and local duality data for non-orientable embedded graphs can be found in [7].

The combinatorial relationship between Eulerian maps and hypermaps on orientable surfaces was studied by Jackson and Visentin [12]. Here we study hyperedge partial duals of hypermaps on arbitrary surfaces.

All hypermaps in this paper are finite. To simplify notation, theorem statements and proofs are given for connected hypermaps; for disconnected hypermaps, the conclusions hold componentwise.

2 Hypermaps and Hyperedge Partial Duals

2.1 The Flag Model

Definition 2.1. A *hypermap* is a triple

$$\mathcal{H} = (X; \tau_0, \tau_1, \tau_2),$$

where X is a finite set, each τ_i is a fixed-point-free involution on X, and the group $\langle \tau_0, \tau_1, \tau_2 \rangle$ generated by these involutions acts transitively on X. The elements of X are called *flags*. For $i, j \in \{0, 1, 2\}$, let

$$X/\langle \tau_i, \tau_j \rangle$$

denote the set of orbits of the subgroup $\langle \tau_i, \tau_j \rangle$ generated by τ_i and τ_j. The sets of vertices, hyperedges, and faces of $\mathcal{H}$ are defined by

$$\mathcal{V}(\mathcal{H}) = X/\langle \tau_1, \tau_2 \rangle, \qquad \mathcal{E}(\mathcal{H}) = X/\langle \tau_0, \tau_2 \rangle, \qquad \mathcal{F}(\mathcal{H}) = X/\langle \tau_0, \tau_1 \rangle,$$

respectively.

Geometrically, a *flag* is a small triangle in the barycentric subdivision of the cell complex associated with $\mathcal{H}$. The involution τ_i crosses the colour-i line, as shown in Figure 1, where blue, red, and black represent 0, 1, and 2, respectively. Thus, at a flag (v, e, f), the involutions τ_0, τ_1, and τ_2 change the vertex, hyperedge, and face components of the flag, respectively, while preserving the other two components. This convention agrees with [2, Section 1.2].

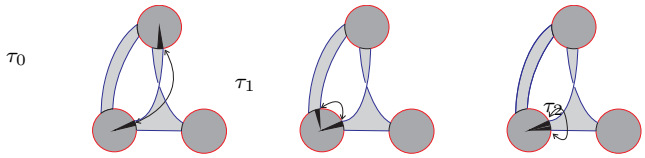


Figure 1: The three basic involutions in the τ-model.

Every orbit of two fixed-point-free involutions is an alternating cycle and hence has even cardinality. For $v \in \mathcal{V}(\mathcal{H})$, $e \in \mathcal{E}(\mathcal{H})$, and $f \in \mathcal{F}(\mathcal{H})$, define the corresponding valences by

$$\mathrm{val}(v) = \frac{|v|}{2}, \qquad \mathrm{val}(e) = \frac{|e|}{2}, \qquad \mathrm{val}(f) = \frac{|f|}{2}.$$

The hypermap $\mathcal{H}$ is called *Eulerian* if every vertex has even valence.

We fix the following graph-theoretic definition of bipartiteness. The *vertex-adjacency multigraph* $\Gamma(\mathcal{H})$ has vertex set $\mathcal{V}(\mathcal{H})$; for every τ_0-transposition $\{x, \tau_0 x\}$, it has an edge whose endpoints are the vertex orbits containing x and $\tau_0 x$. Loops and multiple edges are allowed.

Definition 2.2. The hypermap $\mathcal{H}$ is called *bipartite* if $\Gamma(\mathcal{H})$ is bipartite. This agrees with the usual convention for bipartite hypermaps; see [20, 5]. Equivalently, there is a map $\chi : \mathcal{V}(\mathcal{H}) \to \{+1, -1\}$ such that

$$\chi(V(x)) = -\chi(V(\tau_0 x)) \qquad (x \in X),$$

where $V(x)$ denotes the $\langle \tau_1, \tau_2 \rangle$-orbit containing x.

For an ordinary map, every geometric edge is represented here by two parallel adjacency edges. This does not affect bipartiteness. In the orientable (σ, α)-model, a τ_0-transposition is exactly the adjacency $v(i)$–$v(\alpha(i))$ used in the usual definition of a bipartite hypermap.

2.2 Hyperedge Partial Duality

Let $A \subseteq \mathcal{E}(\mathcal{H})$ be a set of hyperedges, and let

$$X_A = \bigcup_{e \in A} e$$

be the union of their flag sets. Since X_A is invariant under τ_0 and τ_2, the following definition is unambiguous.

Definition 2.3. The *partial dual* of $\mathcal{H} = (X; \tau_0, \tau_1, \tau_2)$ with respect to A is

$$\mathcal{H}^A = (X; \tau_0^A, \tau_1, \tau_2^A),$$

where, for every $x \in X$,

$$\tau_0^A(x) = \begin{cases} \tau_2(x), & x \in X_A, \\ \tau_0(x), & x \notin X_A, \end{cases} \qquad \tau_2^A(x) = \begin{cases} \tau_0(x), & x \in X_A, \\ \tau_2(x), & x \notin X_A. \end{cases}$$

Lemma 2.4. *For every $A \subseteq \mathcal{E}(\mathcal{H})$, the triple*

$$\mathcal{H}^A = (X; \tau_0^A, \tau_1, \tau_2^A)$$

is again a connected hypermap.

Proof. Since X_A is a union of $\langle \tau_0, \tau_2 \rangle$-orbits, both it and its complement $X \setminus X_A$ are invariant under τ_0 and τ_2. Hence, on each of X_A and $X \setminus X_A$, the restrictions of τ_0^A and τ_2^A are just the restrictions of the original involutions τ_0 and τ_2, possibly with their names interchanged. Thus τ_0^A and τ_2^A are fixed-point-free involutions.

View the transpositions of the three involutions as edges of colours $0, 1, 2$ in the flag graph. For every $x \in X$,

$$\{\tau_0^A(x), \tau_2^A(x)\} = \{\tau_0(x), \tau_2(x)\}.$$

Thus the flag graphs of $\mathcal{H}^A$ and $\mathcal{H}$ have the same uncoloured underlying graph: the partial dual only swaps colours 0 and 2 inside the 02-bubbles belonging to X_A, while colour-1 edges remain unchanged. Since the original flag graph is connected, so is the new one. Consequently $\langle \tau_0^A, \tau_1, \tau_2^A \rangle$ acts transitively on X, and $\mathcal{H}^A$ is a connected hypermap. □

Thus the partial dual swaps colours 0 and 2 precisely in the selected 02-bubbles and leaves colour 1 unchanged. This is the hyperedge version of the τ-model formula of Chmutov and Vignes-Tourneret; the direct flag formula is [2, Theorem 2.6], and the colour-swapping formulation in the coloured flag graph is [2, Theorem 2.8]. The pointwise form in Definition 2.3 avoids ambiguities caused by different conventions for permutation composition.

Proposition 2.5. *For any $A, B \subseteq \mathcal{E}(\mathcal{H})$, the following hold:*

(i) *The hyperedge orbits of $\mathcal{H}^A$ are naturally the same as those of $\mathcal{H}$, and hyperedge valences are preserved;*

(ii) $(\mathcal{H}^A)^B = \mathcal{H}^{A \triangle B}$*; in particular,* $(\mathcal{H}^A)^A = \mathcal{H}$.

Proof. On each hyperedge orbit, Definition 2.3 either leaves the pair (τ_0, τ_2) unchanged or swaps its two members. This proves (i). Performing the same swap twice cancels it, giving (ii). □

3 Medial Maps and States

We fix the following running example for the remainder of the paper.

Example 3.1. Let $X = \{1, 2, \ldots, 16\}$, and define

$$\begin{aligned} \tau_0 &= (1\ 8)(2\ 3)(4\ 5)(6\ 7)(9\ 12)(10\ 11)(13\ 16)(14\ 15), \\ \tau_1 &= (1\ 14)(2\ 13)(3\ 12)(4\ 8)(5\ 10)(6\ 16)(7\ 11)(9\ 15), \\ \tau_2 &= (1\ 2)(3\ 4)(5\ 6)(7\ 8)(9\ 10)(11\ 12)(13\ 14)(15\ 16). \end{aligned}$$

Denote the resulting hypermap by $\mathcal{H}_0 = (X; \tau_0, \tau_1, \tau_2)$. Its three vertices are represented by the cyclic orders

$$v = (1, 2, 13, 14), \qquad u = (3, 4, 8, 7, 11, 12), \qquad w = (5, 6, 16, 15, 9, 10),$$

its three hyperedges are

$$h = \{1, 2, 3, 4, 5, 6, 7, 8\}, \qquad a = \{9, 10, 11, 12\}, \qquad b = \{13, 14, 15, 16\},$$

and its unique face is X. The cycles on the vertex discs are read counterclockwise; hence the three discs in Figure 2 correspond to v, u, and w, with the cyclic orders listed above. The valences of h, a, and b are 4, 2, and 2, respectively.

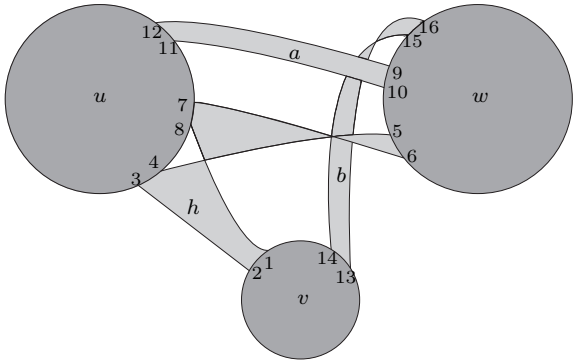


Figure 2: The hypermap $\mathcal{H}_0$.

3.1 The Medial Map in the τ-Model

We now construct the medial map entirely within the τ-model. It is important to distinguish the flag set of the original hypermap from the flag set of the medial map itself: the former becomes the label set of medial half-edges in the latter, but the two sets are not identical.

Let b and w be two symbols, and let $\bar{c}$ denote the symbol different from $c \in \{\mathsf{b}, \mathsf{w}\}$. Set

$$Y = X \times \{\mathsf{b}, \mathsf{w}\}.$$

Write $Y_{\mathsf{b}} = X \times \{\mathsf{b}\}$ and $Y_{\mathsf{w}} = X \times \{\mathsf{w}\}$. Define three involutions on Y by

$$\begin{aligned} \mu_0(x, c) &= (\tau_1 x, c), \\ \mu_2(x, c) &= (x, \bar{c}), \\ \mu_1(x, \mathsf{b}) &= (\tau_2 x, \mathsf{b}), \\ \mu_1(x, \mathsf{w}) &= (\tau_0 x, \mathsf{w}). \end{aligned} \tag{M}$$

Definition 3.2. Let $\mathcal{H} = (X; \tau_0, \tau_1, \tau_2)$ be a hypermap. Its *τ-model representation of the medial map* is defined by

$$M_\tau(\mathcal{H}) = (Y; \mu_0, \mu_1, \mu_2),$$

where Y and the μ_i are given by (M). In what follows we write $M(\mathcal{H})$ for $M_\tau(\mathcal{H})$.

Example 3.3. The medial map $M(\mathcal{H}_0)$ of the hypermap $\mathcal{H}_0$ has three medial vertices v_h, v_a, and v_b, corresponding to the hyperedges h, a, and b, whose counterclockwise half-edge orders are

$$v_h = (\bar{1}, \bar{2}, \bar{3}, \bar{4}, \bar{5}, \bar{6}, \bar{7}, \bar{8}), \qquad v_a = (\bar{9}, \bar{10}, \bar{11}, \bar{12}), \qquad v_b = (\bar{13}, \bar{14}, \bar{15}, \bar{16}),$$

where $\bar{x} = \{x_{\mathsf{b}}, x_{\mathsf{w}}\}$. It has eight medial edges

$$m_{1,14},\ m_{2,13},\ m_{3,12},\ m_{4,8},\ m_{5,10},\ m_{6,16},\ m_{7,11},\ m_{9,15},$$

where $m_{4,8}$ is a loop at v_h; $m_{1,14}$, $m_{2,13}$, and $m_{6,16}$ join v_h to v_b; $m_{3,12}$, $m_{5,10}$, and $m_{7,11}$ join v_h to v_a; and $m_{9,15}$ joins v_a to v_b. Figure 3 depicts the three medial vertices and eight medial edges in this order.

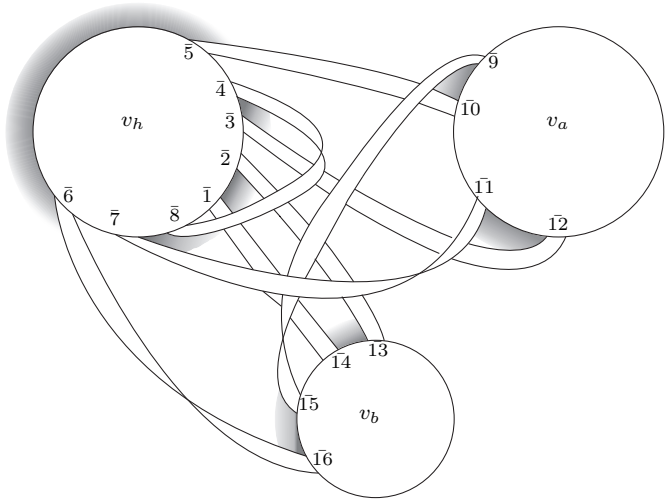


Figure 3: The medial map $M(\mathcal{H}_0)$ of $\mathcal{H}_0$, with the canonical checkerboard coloring around its vertices.

Lemma 3.4. *Let $Y/\langle\mu_2\rangle$ denote the set of orbits of $\langle\mu_2\rangle$. Then the map*

$$\iota : X \longrightarrow Y/\langle\mu_2\rangle, \qquad \iota(x) = [(x, \mathsf{b})]_{\langle\mu_2\rangle} = \{(x, \mathsf{b}), (x, \mathsf{w})\}$$

is a bijection.

More precisely, let $\mathrm{Orb}_{\langle\tau,\tau'\rangle}(x)$ be the orbit of the flag x under the group generated by τ and τ'.

Then

$$\mathrm{Orb}_{\langle\mu_1,\mu_2\rangle}(x, \mathsf{b}) = e \times \{\mathsf{b}, \mathsf{w}\}, \tag{1}$$
$$\mathrm{Orb}_{\langle\mu_0,\mu_2\rangle}(x, \mathsf{b}) = \{(x, \mathsf{b}), (x, \mathsf{w}), (\tau_1 x, \mathsf{b}), (\tau_1 x, \mathsf{w})\}, \tag{2}$$

where $e = \mathrm{Orb}_{\langle\tau_0,\tau_2\rangle}(x)$.

Proof. Since $\mu_2(x, \mathsf{b}) = (x, \mathsf{w})$ and μ_2 is fixed-point-free, ι is plainly a bijection.

From

$$\mu_1(x, \mathsf{b}) = (\tau_2 x, \mathsf{b}), \qquad \mu_2\mu_1(x, \mathsf{b}) = (\tau_2 x, \mathsf{w}), \qquad \mu_1\mu_2(x, \mathsf{b}) = (\tau_0 x, \mathsf{w}), \qquad \mu_2\mu_1\mu_2(x, \mathsf{b}) = (\tau_0 x, \mathsf{b})$$

the $\langle \mu_1, \mu_2 \rangle$-orbit contains $e \times \{\mathsf{b}, \mathsf{w}\}$. Conversely, μ_1 applies only τ_2 on the b-layer and only τ_0 on the w-layer, while μ_2 does not change the first coordinate. Thus the orbit cannot leave $e \times \{\mathsf{b}, \mathsf{w}\}$, proving (1).

Moreover, μ_0 and μ_2 commute, and μ_0 applies τ_1 to the first coordinate, so (2) follows. □

Proposition 3.5. *The $M(\mathcal{H})$ defined in Definition 3.2 is a map; that is, every $\langle \mu_0, \mu_2 \rangle$-orbit contains exactly four flags. Its cells have the following natural correspondence with the data of $\mathcal{H}$:*

The τ-data of $\mathcal{H}$	*Object in $M(\mathcal{H})$*
$e \in X/\langle \tau_0, \tau_2 \rangle$	*a medial vertex v_e*
$x \in e$	*a medial half-edge at v_e labelled by x*
$\{x, \tau_1 x\}$	*a medial edge*
$X/\langle \tau_1, \tau_2 \rangle$	*a black face*
$X/\langle \tau_1, \tau_0 \rangle$	*a white face.*

In particular, if e has valence k, then v_e has degree $2k$. At v_e, the black angles project to pairs $\{x, \tau_2 x\}$ and the white angles to pairs $\{x, \tau_0 x\}$. Thus $M(\mathcal{H})$ has the canonical checkerboard colouring.

Proof. The three maps μ_i are fixed-point-free involutions. Starting with a flag (x, b), the maps μ_0, μ_1, and $\mu_2\mu_1\mu_2$ act on the first coordinate by τ_1, τ_2, and τ_0, respectively, while μ_2 changes the second coordinate. Hence the connectedness of $\mathcal{H}$ implies that $M(\mathcal{H})$ is connected. Also, μ_0 and μ_2 commute. By (2), every edge of $M(\mathcal{H})$ consists of four flags, so $M(\mathcal{H})$ is a map; moreover, its edges are in bijection with the τ_1-transpositions. By (1),

$$Y/\langle \mu_1, \mu_2 \rangle \cong X/\langle \tau_0, \tau_2 \rangle.$$

This gives the correspondence between the medial vertices and the hyperedges of the original hypermap. Since $M(\mathcal{H})$ has been shown to be a map, μ_2 preserves its vertices and edges while interchanging the two face-side flags; hence $Y/\langle \mu_2 \rangle$ is precisely its set of half-edges. By Lemma 3.4, $\iota : X \to \mathcal{I}(M(\mathcal{H}))$ is a bijection, and the half-edge set at v_e is precisely $\{\iota(z) : z \in e\}$. Therefore, if $\mathrm{val}_{\mathcal{H}}(e) = k$, then

$$\deg_{M(\mathcal{H})}(v_e) = |e| = 2k.$$

Since μ_0 preserves the second coordinate, whereas μ_1 acts by τ_2 on the b-layer and by τ_0 on the w-layer,

$$Y_{\mathsf{b}}/\langle \mu_0, \mu_1 \rangle \cong X/\langle \tau_1, \tau_2 \rangle,$$

and

$$Y_{\mathsf{w}}/\langle \mu_0, \mu_1 \rangle \cong X/\langle \tau_1, \tau_0 \rangle.$$

Thus the former is a black face and the latter a white face. At a medial vertex, a black face turns according to the action of μ_1 on the b-layer, hence via a τ_2-pair after projection; the white case is analogous, with τ_0. This proves the assertions about angles and the checkerboard colouring. □

Notice that, when the hypermap is orientable, the construction of the medial map given here coincides with the permutation construction of Cori–Hetyei [4, Definition 3.1].

The bijection ι of Lemma 3.4 says precisely that, after forgetting the second coordinate of Y, X is the label set of the medial half-edges of $M(\mathcal{H})$. Geometrically, in the flag triangulation of $\mathcal{H}$, take a small closed neighbourhood of each 02-cell as a medial vertex disc and attach a narrow medial band across every colour-1 flag-graph edge $\{x, \tau_1 x\}$; its two ends are labelled by x and $\tau_1 x$. Its black and white sides turn at vertex discs according to the τ_2- and τ_0-pairings, respectively, so the flag involutions of the resulting ribbon graph are exactly (M). Thus $M(\mathcal{H})$ is not merely a combinatorial map but the geometric medial map of $\mathcal{H}$. In the non-orientable case, an 02-cycle provides only an unoriented cyclic order; the local orientation systems below record how bands are glued relative to the vertex discs.

3.2 States

Recall from Lemma 3.4 that every flag of the original hypermap corresponds to one medial half-edge. Although the state circles are drawn on the medial map, we describe them using the flag set X through this bijection.

The involutions μ_0, μ_1, μ_2 act on the flag set Y of the medial map, whereas τ_0, τ_1, τ_2 act on the flag set X of the original hypermap. Under the identification given by ι, a τ_1-transposition is a medial edge. The involutions τ_0 and τ_2 describe the two possible turns at a medial vertex.

We now explain which of these two turns is the black smoothing and which is the white smoothing. By the definition of the medial map, $\mu_1(x, \mathsf{b}) = (\tau_2 x, \mathsf{b}), \mu_1(x, \mathsf{w}) = (\tau_0 x, \mathsf{w})$. The two layers b and w represent the two sides of the medial map. Furthermore, by Proposition 3.5, $Y_{\mathsf{b}}/\langle \mu_0, \mu_1 \rangle \cong X/\langle \tau_1, \tau_2 \rangle$ is the set of black faces, while $Y_{\mathsf{w}}/\langle \mu_0, \mu_1 \rangle \cong X/\langle \tau_1, \tau_0 \rangle$ is the set of white faces.

A smoothing at a medial vertex joins two incident half-edges which are connected through one side of the medial map. On the black side, the connection is determined by τ_2, since μ_1 acts as τ_2 on the b-layer. On the white side, the connection is determined by τ_0, since μ_1 acts as τ_0 on the w-layer. Therefore,

τ_2 corresponds to the black smoothing, whereas τ_0 corresponds to the white smoothing.

Define an involution η_A on X by

$$\eta_A(x) = \begin{cases} \tau_0(x), & x \in X_A, \\ \tau_2(x), & x \notin X_A. \end{cases}$$

The reason for this choice is the definition of hyperedge partial duality. Notice that $\eta_A = \tau_2^A$ as involutions on the original flag set X. Geometrically, this means that the state uses the white smoothing at every hyperedge in A and the black smoothing at every hyperedge outside A.

Definition 3.6. For $A \subseteq \mathcal{E}(\mathcal{H})$, the *state* S_A is the edge-coloured multigraph on X whose two edge classes are the transpositions of τ_1 and the transpositions of η_A.

Using the bijection ι, the same graph may be regarded as a graph on the half-edge set of the medial map. In this interpretation, the τ_1-edges are the medial edges, and the η_A-edges are the chosen smoothing connections at medial vertices. The connected components of S_A are called *state circles*.

If a τ_1-transposition and an η_A-transposition have the same endpoints, they are still regarded as two parallel edges of different colours. For a state circle C, let $E_{\tau_1}(C)$ be the set of all τ_1-edges in C, and define

$$\ell(C) := |E_{\tau_1}(C)|\,.$$

Since both τ_1 and η_A are fixed-point-free, every flag of S_A is incident with exactly one edge of each colour. Hence every connected component of S_A is a cycle with alternating edge colours. We allow $\ell(C) = 1$; in that case, C is a 2-cycle consisting of two parallel edges of different colours.

Theorem 3.7. *The state circles of S_A are naturally in bijection with the vertices of $\mathcal{H}^A$. If C is a state circle and v_C is the corresponding vertex, then*

$$\mathrm{val}_{\mathcal{H}^A}(v_C) = \ell(C).$$

Proof. Since $\tau_2^A = \eta_A$, the vertices of $\mathcal{H}^A$ are exactly the $\langle \tau_1, \eta_A\rangle$-orbits in X. Moreover, because the edges of S_A are precisely the transpositions occurring in τ_1 and η_A, these orbits coincide with the connected components of S_A. Hence the state circles are in natural bijection with the vertices of $\mathcal{H}^A$.

Let C be one such component. Since τ_1 is fixed-point-free, its transpositions partition the flags of C into disjoint pairs. Hence

$$|C| = 2\,|E_{\tau_1}(C)|\,.$$

The flag set of the corresponding vertex v_C is C. Therefore,

$$\mathrm{val}_{\mathcal{H}^A}(v_C) = \frac{|C|}{2} = |E_{\tau_1}(C)| = \ell(C).$$

□

Corollary 3.8. *The partial dual $\mathcal{H}^A$ is Eulerian if and only if every state circle of S_A has even length.*

Proof. By Theorem 3.7, the valence of each vertex of $\mathcal{H}^A$ is equal to the length of the corresponding state circle. Therefore, all vertex valences are even if and only if all state circles have even length. □

4 Eulerian Partial Duals

4.1 Crossing-Total Orientations

By Lemma 3.4, we identify X with the half-edge set of $M(\mathcal{H})$ via ι. Let Ω be an orientation of the edges of $M(\mathcal{H})$. For $x \in X$ (the medial half-edge labelled by x), set

$$\delta_\Omega(x) = \begin{cases} +1, & \text{if the medial edge is directed away from the medial vertex at } x, \\ -1, & \text{if the medial edge is directed towards the medial vertex at } x. \end{cases}$$

Then $\delta_\Omega(\tau_1 x) = -\delta_\Omega(x)$. A black angle is a source or sink precisely when $\delta_\Omega(\tau_2 x) = \delta_\Omega(x)$, and a white angle is a source or sink precisely when $\delta_\Omega(\tau_0 x) = \delta_\Omega(x)$.

Definition 4.1. Let e be a hyperedge, regarded as a medial vertex.

(a) If δ_Ω is constant on e, then e is of *t-type*.

(b) If e is not of t-type and $\delta_\Omega(\tau_2 x) = \delta_\Omega(x)$ for every $x \in e$, then e is of *c-type*.

(c) If e is not of t-type and $\delta_\Omega(\tau_0 x) = \delta_\Omega(x)$ for every $x \in e$, then e is of *d-type*.

If every hyperedge is of one of these three types, then Ω is called a *crossing-total orientation*. Let $D(\Omega)$ and $T(\Omega)$ denote the sets of d-type and t-type hyperedges, respectively.

The three types are pairwise disjoint. Indeed, if the two same-sign conditions in (b) and (c) both hold, the connectedness of the 02-cycle of e implies that δ_Ω is constant on e.

Theorem 4.2. *Let $\mathcal{H}$ be a hypermap and $A \subseteq \mathcal{E}(\mathcal{H})$. Then $\mathcal{H}^A$ is Eulerian if and only if there is a crossing-total orientation Ω of $M(\mathcal{H})$ and a set $T' \subseteq T(\Omega)$ such that*

$$A = D(\Omega) \cup T'.$$

Proof. Suppose first that $\mathcal{H}^A$ is Eulerian. By Corollary 3.8, every state circle C of S_A has even length. Choose a flag x_0 on each state circle and assign it the sign $+1$. Propagate signs according to

$$\delta(\eta_A x) = \delta(x), \qquad \delta(\tau_1 x) = -\delta(x). \tag{E}$$

When one traverses C once, exactly $\ell(C)$ τ_1-edges are crossed, so the cycle relation gives the multiplier $(-1)^{\ell(C)} = 1$; hence the assignment is well-defined.

Each τ_1-transposition $\{x, \tau_1 x\}$ corresponds to a medial edge. Equation (E) gives $\delta(\tau_1 x) = -\delta(x)$. Thus the medial edge has a unique orientation for which it is directed away from the corresponding medial vertex at the endpoint labelled x if and only if $\delta(x) = +1$. This yields an orientation Ω of $M(\mathcal{H})$.

If $e \in A$, then $\eta_A|_e = \tau_0|_e$, so for every $x \in e$,

$$\delta(\tau_0 x) = \delta(x).$$

If δ is constant on e, then e is of t-type; otherwise it is of d-type. If $e \notin A$, then $\eta_A|_e = \tau_2|_e$, so for every $x \in e$,

$$\delta(\tau_2 x) = \delta(x).$$

If δ is constant on e, then e is of t-type; otherwise it is of c-type. Thus Ω is crossing-total. Since the three types are pairwise disjoint,

$$D(\Omega) \subseteq A, \qquad A \setminus D(\Omega) \subseteq T(\Omega).$$

Set

$$T' := A \setminus D(\Omega)$$

Then $T' \subseteq T(\Omega)$ and $A = D(\Omega) \cup T'$.

Conversely, suppose that Ω is crossing-total and

$$A = D(\Omega) \cup T', \qquad T' \subseteq T(\Omega).$$

Let δ_Ω be the endpoint-sign function of this orientation. For $x \in X$, let $e(x)$ be the hyperedge containing x. If $e(x) \in A$, then $e(x)$ belongs either to $D(\Omega)$ or to T'. In the first case $e(x)$ is of d-type and $\eta_A = \tau_0$, while in the second it is of t-type. Thus in both cases

$$\delta_\Omega(\eta_A x) = \delta_\Omega(x).$$

If $e(x) \notin A$, then it does not belong to $D(\Omega)$. Since Ω is crossing-total and the three types are pairwise disjoint, $e(x)$ is of c-type or t-type; now $\eta_A = \tau_2$, so again

$$\delta_\Omega(\eta_A x) = \delta_\Omega(x).$$

On the other hand, every directed medial edge satisfies

$$\delta_\Omega(\tau_1 x) = -\delta_\Omega(x).$$

Now let C be any state circle of S_A. Starting from a flag of C and walking around it, the sign is unchanged across an η_A-edge and changes across a τ_1-edge. Returning to the initial flag after one circuit gives

$$\delta_\Omega(x) = (-1)^{\ell(C)}\delta_\Omega(x).$$

Since $\delta_\Omega(x) \in \{+1, -1\}$, $\ell(C)$ must be even. Hence every state circle of S_A has even length, and Corollary 3.8 implies that $\mathcal{H}^A$ is Eulerian. □

5 Modified Medial Maps and All-Crossing Orientations

The preceding proof does not use a global orientation. Bipartiteness is different: the usual left–right labels of a directed medial edge cannot be propagated consistently along a path that passes through twisted medial bands. We replace these labels by local orientations on vertex discs and the band-gluing type relative to that choice.

5.1 Relative Twist Data

Definition 5.1. A *local orientation system* for $M(\mathcal{H})$ is a map $\omega : X \to \{+1, -1\}$ satisfying

$$\omega(\tau_0 x) = -\omega(x), \qquad \omega(\tau_2 x) = -\omega(x) \qquad (x \in X).$$

Such a system always exists: choose a sign on each 02-orbit and propagate it around the alternating cycle. On a fixed hyperedge there are two choices, differing only by a common sign.

Lemma 5.2. *For any local orientation system ω, define $q_\omega : Y \to \{+1, -1\}$ by*

$$q_\omega(x, \mathsf{b}) = \omega(x), \qquad q_\omega(x, \mathsf{w}) = -\omega(x).$$

Then

$$q_\omega(\mu_1 y) = -q_\omega(y), \qquad q_\omega(\mu_2 y) = -q_\omega(y) \qquad (y \in Y). \tag{L}$$

Conversely, every map $q : Y \to \{+1, -1\}$ satisfying (L) *arises uniquely from a local orientation system.*

In the flag triangulation of $M(\mathcal{H})$, q_ω is precisely the flag-sign representation of the local orientation of each medial vertex disc: on every $\langle \mu_1, \mu_2 \rangle$-orbit, the restriction of q_ω and its negative correspond to the two local orientations of that vertex disc.

Proof. The second equality follows immediately from $\mu_2(x, \mathsf{c}) = (x, \bar{\mathsf{c}})$, where $\{\mathsf{c}, \bar{\mathsf{c}}\} = \{\mathsf{b}, \mathsf{w}\}$. If $y = (x, \mathsf{b})$, then

$$q_\omega(\mu_1 y) = q_\omega(\tau_2 x, \mathsf{b}) = \omega(\tau_2 x) = -\omega(x) = -q_\omega(y).$$

If $y = (x, \mathsf{w})$, then

$$q_\omega(\mu_1 y) = q_\omega(\tau_0 x, \mathsf{w}) = -\omega(\tau_0 x) = \omega(x) = -q_\omega(y).$$

Thus (L) holds.

Conversely, let $\omega(x) = q(x, \mathsf{b})$. The μ_2 relation gives $q(x, \mathsf{w}) = -\omega(x)$; applying the μ_1 relation at (x, b) and (x, w), respectively, gives

$$\omega(\tau_2 x) = -\omega(x), \qquad \omega(\tau_0 x) = -\omega(x).$$

Thus ω is a local orientation system, and uniqueness is clear.

Finally, the flags of a medial vertex disc form exactly its $\langle \mu_1, \mu_2 \rangle$-orbit, whose flag triangulation is a disc. Adjacent small triangles meet across colour-1 or colour-2 edges; with the colour order fixed, an orientation of the disc is exactly one for which adjacent triangles have opposite flag signs. Conversely, the alternating signs in (L) make the orientations of these triangles compatible; since the complex is a disc, they determine a unique local orientation. Changing all signs reverses the orientation of the disc. □

For a medial edge $a = \{x, \tau_1 x\}$, define its *twist indicator* relative to ω by $\varepsilon_\omega(a) \in \{0, 1\}$,

$$\omega(\tau_1 x) = (-1)^{1+\varepsilon_\omega(a)} \omega(x). \tag{T}$$

This definition is independent of the choice of endpoint x: exchanging x and $\tau_1 x$ replaces the ratio of the two values in $\{+1, -1\}$ by its reciprocal, which is the same ratio.

Lemma 5.3. *Let $a = \{x, \tau_1 x\}$ be a medial edge. Then for $c \in \{\mathsf{b}, \mathsf{w}\}$,*

$$q_\omega\big(\mu_0(x, c)\big) = (-1)^{1+\varepsilon_\omega(a)} q_\omega(x, c). \tag{U}$$

Thus, relative to the local orientations encoded by q_ω, $\varepsilon_\omega(a) = 0$ if and only if the band a is untwisted, while $\varepsilon_\omega(a) = 1$ if and only if the band is twisted.

If ω' is another local orientation system and

$$\omega'(x) = r(e(x))\omega(x), \qquad r(e) = (-1)^{\rho(e)},$$

where $e(x)$ is the hyperedge containing x, then

$$\varepsilon_{\omega'}(a) \equiv \varepsilon_\omega(a) + \rho(e(x)) + \rho(e(\tau_1 x)) \pmod 2. \tag{S}$$

Proof. For $c = \mathsf{b}$,

$$q_\omega(\mu_0(x, \mathsf{b})) = q_\omega(\tau_1 x, \mathsf{b}) = \omega(\tau_1 x) = (-1)^{1+\varepsilon_\omega(a)} q_\omega(x, \mathsf{b}).$$

For $c = \mathsf{w}$, both sides acquire an additional minus sign, so the same equality remains valid.

Represent the band by a rectangle. The flags (x, c) and $\mu_0(x, c) = (\tau_1 x, c)$ lie at the two ends on the same side of the rectangle. With the colour order fixed, the local orientations of the endpoint discs extend compatibly across the rectangle if and only if the two signs across this colour-0 flag edge are opposite; this is exactly

$$q_\omega(\mu_0(x, c)) = -q_\omega(x, c).$$

If the two signs agree, transporting a local orientation along the band produces the opposite orientation, so the band is twisted relative to the chosen local orientations (with twisting taken modulo 2). Together with (U), this proves the criterion. The criterion is independent of the choice of c.

For $y = (x, c)$ we also have

$$q_{\omega'}(y) = r(e(x)) q_\omega(y).$$

Thus

$$\begin{aligned} q_{\omega'}(\mu_0 y) &= r(e(\tau_1 x)) q_\omega(\mu_0 y) \\ &= (-1)^{1+\varepsilon_\omega(a)} r(e(\tau_1 x)) r(e(x)) q_{\omega'}(y). \end{aligned}$$

Substituting $r = (-1)^\rho$ and comparing with (U) gives (S).

If a is a loop, then $e(x) = e(\tau_1 x)$ and the additional term in the formula is $2\rho(e(x)) \equiv 0 \pmod 2$. Thus reversing the same vertex disc reverses both endpoint local orientations but does not change the twist of a loop band, in agreement with its topological definition. □

Lemma 5.3 shows that the twist indicator is not an intrinsic 0–1 label: when the local orientation of a medial vertex disc is changed, the band indicators transform according to (S). This is exactly the switching operation in a signed rotation representation. Thus the underlying medial map is determined by the τ-model, while $(\omega, \varepsilon_\omega)$ is only one local signed description of it.

Definition 5.4. For a local orientation system ω, the *modified medial map* $\widehat{M_\omega}(\mathcal{H})$ is obtained from $M(\mathcal{H})$ by inserting an artificial bivalent bar into each medial edge a satisfying $\varepsilon_\omega(a) = 1$. The checkerboard colouring of $M(\mathcal{H})$ extends across every subdivision.

Figure 4 illustrates this local operation: when a medial band is twisted relative to the chosen local orientations, an artificial bivalent bar is inserted in its interior, splitting the original edge into two segments. This artificial bar represents neither a hyperedge of the original hypermap nor an original medial vertex; it records only the twisting obstruction of the band. In the orientation definitions below, this bivalent bar is required to be a source or a sink.

Thus the set of subdivided medial edges generally depends on ω. The following lemma shows, however, that the existence of an all-crossing orientation relevant to bipartite partial duals, and the associated set $C(\Phi)$, do not depend on this local choice.

The following lemma provides a convenient algebraic encoding of such orientations.



Figure 4: Local construction of the modified medial map.

Lemma 5.5. *Orientations of $\widehat{M}_\omega(\mathcal{H})$ for which every artificial bivalent bar is a source or a sink are in bijection with maps $\delta : X \to \{+1, -1\}$ satisfying*

$$\delta(\tau_1 x) = (-1)^{1+\varepsilon_\omega(\{x,\tau_1 x\})}\delta(x) \qquad (x \in X). \tag{D}$$

Here $\delta(x) = +1$ means that the edge segment at x is directed away from its incident original medial vertex.

Proof. If $\varepsilon_\omega(a) = 0$, the edge a is not subdivided and its endpoint signs are opposite, as for an ordinary directed edge. If $\varepsilon_\omega(a) = 1$, the two segments meet at an artificial source or sink, and their signs at the two original medial vertices agree. These two local cases give exactly (D), and the construction is reversible. □

Definition 5.6. Let Φ be an orientation encoded by δ as in Lemma 5.5. If, for every $x \in e$,

$$\delta(\tau_2 x) = \delta(x), \qquad \delta(\tau_0 x) = -\delta(x),$$

then e is called a *c-type all-crossing vertex*; if, for every $x \in e$,

$$\delta(\tau_0 x) = \delta(x), \qquad \delta(\tau_2 x) = -\delta(x),$$

then it is called a *d-type all-crossing vertex*. If every original medial vertex is of exactly one of these two types, then Φ is called *all-crossing*. Let $C(\Phi)$ denote the set of its c-type hyperedges.

Thus, at a c-type vertex all black angles are sources or sinks and all white angles are crossings; at a d-type vertex the roles of black and white are interchanged.

To connect these definitions with the running example, Figure 5 shows an all-crossing orientation on the medial map in Figure 3. The four small discs in the figure lie inside the twisted medial edges $m_{4,8}$, $m_{7,11}$, $m_{6,16}$, and $m_{9,15}$, and therefore represent the four artificial bivalent bars of $\widehat{M}_\omega(\mathcal{H}_0)$. The arrows give the directions of the edge segments and form a source or sink at each artificial bar. In the half-edge orders at v_h, v_a, v_b, the endpoint signs are

$$\delta = (+,+,-,-,+,+,-,-;\ +,-,-,+;\ -,-,+,+).$$

Thus h and b are c-type hyperedges and a is d-type, so the set of c-type hyperedges is $C(\Phi) = \{h, b\}$. This gives a concrete example of the set $A = C(\Phi)$ in Theorem 6.1.

Remark 5.7. If a hyperedge e has valence k, then the corresponding medial vertex v_e has degree $2k$. Definition 4.1 uses the three types c, d, t, whereas Definition 5.6 uses only c, d. When $k > 2$, our c- and d-types are definitions for higher-valence medial vertices: c-type requires every black angle to be a

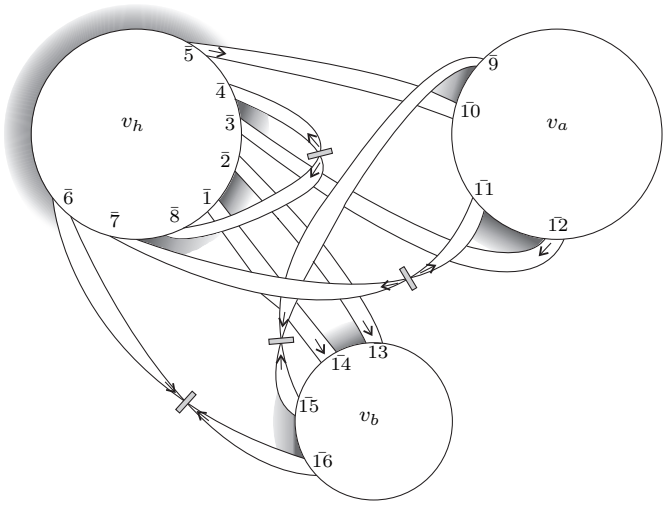


Figure 5: An all-crossing orientation of $\mathcal{H}_0$.

source or sink and every white angle to be a crossing, while d-type interchanges the roles of black and white. We do not apply the four-valent "in,in,out,out" pattern to higher-valence vertices without qualification. Only when $k = 2$ is v_e four-valent, and our c, d, t types are exactly the classical local patterns in a medial graph.

Lemma 5.8. *The existence of an all-crossing orientation with a prescribed set $C(\Phi)$ is independent of the choice of local orientation system ω.*

Proof. Let ω' be another local orientation system. Since the ratio ω'/ω is invariant under both τ_0 and τ_2, it is constant on each hyperedge. Thus there is a map $r : \mathcal{E}(\mathcal{H}) \to \{+1, -1\}$ such that $\omega'(x) = r(e(x))\omega(x)$, where $e(x)$ is the hyperedge containing x. If δ satisfies (D), set $\delta'(x) = r(e(x))\delta(x)$. For $a = \{x, \tau_1 x\}$, equations (D) and (S) give

$$\begin{aligned}\delta'(\tau_1 x) &= r(e(\tau_1 x))(-1)^{1+\varepsilon_\omega(a)}\delta(x) \\ &= (-1)^{1+\varepsilon_{\omega'}(a)} r(e(x))\delta(x) \\ &= (-1)^{1+\varepsilon_{\omega'}(a)}\delta'(x).\end{aligned}$$

Thus δ' satisfies the corresponding form of (D) for ω'. Applying the same transformation with r recovers δ, so this gives a bijection between the two classes of orientations. For a pair of τ_0- or τ_2-paired flags, both flags lie in the same hyperedge and are multiplied by the same factor $r(e)$. Consequently all same-sign and opposite-sign conditions in Definition 5.6 are preserved, and in particular $C(\Phi)$ is unchanged. □

6 Bipartite Partial Duals

Theorem 6.1. *Let $\mathcal{H}$ be a hypermap, and let $A \subseteq \mathcal{E}(\mathcal{H})$. Then the partial dual $\mathcal{H}^A$ is bipartite if and only if, for any local orientation system ω, the modified*

medial map $\widehat{M}_\omega(\mathcal{H})$ admits an all-crossing orientation Φ such that

$$A = C(\Phi).$$

Proof. By Lemma 5.8, this condition is independent of the chosen local orientation system ω.

Suppose first that $\mathcal{H}^A$ is bipartite. Let $\lambda : X \to \{+1, -1\}$ be the pullback of a bipartition of its vertex-adjacency multigraph. Since the vertices of $\mathcal{H}^A$ are the $\langle \tau_1, \tau_2^A \rangle$-orbits,

$$\lambda(\tau_1 x) = \lambda(\tau_2^A x) = \lambda(x), \qquad \lambda(\tau_0^A x) = -\lambda(x). \tag{B}$$

Set $\delta(x) = \lambda(x)\omega(x)$. The first equality in (B), together with (T), implies (D); hence δ gives an orientation of the modified medial map.

If $e \in A$, then on e we have $\tau_2^A = \tau_0$ and $\tau_0^A = \tau_2$. Using the identities in (B) and the fact that ω changes sign under both τ_0 and τ_2, we obtain

$$\delta(\tau_0 x) = -\delta(x), \qquad \delta(\tau_2 x) = \delta(x) \qquad (x \in e).$$

Thus e is of c-type. If $e \notin A$, the same calculation gives

$$\delta(\tau_0 x) = \delta(x), \qquad \delta(\tau_2 x) = -\delta(x) \qquad (x \in e),$$

so e is of d-type. The resulting orientation is therefore all-crossing, and $C(\Phi) = A$.

Conversely, let Φ be an all-crossing orientation with $C(\Phi) = A$, and let δ be its signed direction map. Define

$$\lambda(x) = \delta(x)\omega(x).$$

Equations (T) and (D) show that $\lambda(\tau_1 x) = \lambda(x)$. If $e \in A$, the c-type conditions give

$$\lambda(\tau_0 x) = \lambda(x), \qquad \lambda(\tau_2 x) = -\lambda(x).$$

Since $\tau_2^A = \tau_0$ and $\tau_0^A = \tau_2$ on e, this yields

$$\lambda(\tau_2^A x) = \lambda(x), \qquad \lambda(\tau_0^A x) = -\lambda(x).$$

If $e \notin A$, the d-type conditions give the same two identities, since now $\tau_2^A = \tau_2$ and $\tau_0^A = \tau_0$. Hence λ is constant on every vertex orbit of $\mathcal{H}^A$ and changes sign along every adjacency edge of $\Gamma(\mathcal{H}^A)$. It is therefore a valid bipartition of $\mathcal{H}^A$. □

Example 6.2. It is easy to check that $\mathcal{H}_0$ has exactly one all-crossing orientation, as shown in Figure 5. Let $A = \{h, b\}$ denote the set of c-type edges identified by the all-crossing orientation. The identity $\mathcal{V}(\mathcal{H}_0^A) = X/\langle \tau_1, \tau_2^A \rangle$ directly yields the two vertex orbits

$$P = (1, 4, 5, 8, 9, 10, 14, 15), \qquad Q = (2, 3, 6, 7, 11, 12, 13, 16).$$

In $\Gamma(\mathcal{H}_0^A)$, every adjacency edge contributed by h, a, and b joins P to Q. Hence $\{P\} \sqcup \{Q\}$ is a bipartition, so $\mathcal{H}_0^A$ is the bipartite partial dual of $\mathcal{H}_0$.

Corollary 6.3. *If $\mathcal{H}^A$ is bipartite for some $A \subseteq \mathcal{E}(\mathcal{H})$, then every hyperedge of $\mathcal{H}$ has even valence.*

Proof. By Theorem 6.1, every hyperedge in an all-crossing orientation is of c- or d-type. Around a c-type hyperedge, the sign δ changes across every τ_0-pair and is unchanged across every τ_2-pair. Thus, on traversing the alternating 02-cycle once, the sign changes once for every τ_0-pair. It must return to its initial value, so the number of such pairs, namely the valence of the hyperedge, is even. The d-type case is identical with the roles of τ_0 and τ_2 exchanged. □

Example 6.4. The projective-plane hypermap depicted in Figure 1 has a hyperedge of valence three. Corollary 6.3 immediately implies that neither of its four hyperedge partial duals is bipartite.

Remark 6.5. In the following two special cases, our constructions reduce to existing orientation characterizations.

(i) If $\mathcal{H}$ is orientable, choose a global flag-sign function ω such that

$$\omega(\tau_i x) = -\omega(x) \qquad (i = 0, 1, 2).$$

Then $\varepsilon_\omega(a) = 0$ for every medial edge a, and hence $\widehat{M}_\omega(\mathcal{H}) = M(\mathcal{H})$. Under the standard correspondence between the τ- and (σ, α)-models (see [2, Section 1.3 and Theorem 2.4]; the permutation construction of the medial map is in [4, Definition 3.1]), Theorems 4.2 and 6.1 reduce respectively to [10, Theorems 5.3 and 6.5].

(ii) If every hyperedge e satisfies $\mathrm{val}(e) = 2$, then every $\langle \tau_0, \tau_2 \rangle$-orbit has four flags, so $\mathcal{H}$ is a map and determines a ribbon graph G. Under the natural correspondence between hyperedges and edges of G, $M(\mathcal{H})$ is isomorphic, as a checkerboard-coloured map, to the usual medial graph G_m of G; every v_e is four-valent. In this case, t-type means that all four incident half-edges point in or all point out, while c- and d-types are the two classical all-crossing patterns in which the black or white angles are sources/sinks and the other colour consists of crossings.

Furthermore, by Definition 2.3, under the natural correspondence between hyperedges and edges of G, $\mathcal{H}^A$ is exactly the ribbon-graph partial dual G^A: both swap colours 0 and 2 in the four-flag 02-orbit of each selected edge. By Lemma 5.3, ω is a choice of local orientations for the vertex discs of G_m, and $\varepsilon_\omega(a) = 1$ precisely when a is twisted relative to this choice. Thus $\widehat{M}_\omega(\mathcal{H})$ is isomorphic to the modified medial graph $\widehat{G}_m$ obtained from the same local orientation system. Theorem 6.1 therefore reduces to [6, Theorem 1.4], and Theorem 4.2 to [6, Theorem 1.5].

Proof. For (i), orientability is equivalent to a bipartition of the flag graph in which every τ_i changes the sign; see [2, Section 1.3]. Equation (T) then gives $\varepsilon_\omega = 0$. For (ii), the four-flag 02-orbit is exactly the criterion for a map; the flag correspondence for the medial map is given by Proposition 3.5, and Lemma 5.3 identifies band twists with the modified medial graph under the same local orientations. The remaining assertions about local types follow immediately from the alternating black and white angles at a four-valent vertex. □

7 Conclusion

The τ-model renders both characterizations independent of orientability. The state-circle argument identifies the common mechanism behind the Eulerian theorem, while local orientation systems encode precisely the extra data needed for bipartiteness. Because vertices, edges, and faces play symmetric roles in a hypermap, the analogous characterizations for vertex and face partial duals are obtained simply by interchanging the subscripts of τ_i.

Declaration of competing interest

We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work.

Acknowledgements

This work is supported by National Natural Science Foundation of China (Grant Number: 12661073) and Natural Science Foundation of Xinjiang (Grant Number: 2024D01A89).

Data availability

No data was used for the research described in the article.

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